AI-Powered RAG System (Document Assistant)

Full System Architecture & Application Flow

Technology Stack

Framework

- .NET 8 Web API
- ASP.NET Core
- Semantic Kernel
- Entity Framework Core
- SignalR (Realtime Progress)
- MediatR (CQRS)
- FluentValidation
- Serilog

AI

- OpenAI GPT-4.1 / GPT-4o
- Google Gemini 2.x
- Semantic Kernel

Embedding Models

- text-embedding-3-small
- text-embedding-3-large
- Gemini Embeddings

Vector Database

- Pinecone
- Milvus
- Qdrant (Optional Local)

Database

- PostgreSQL

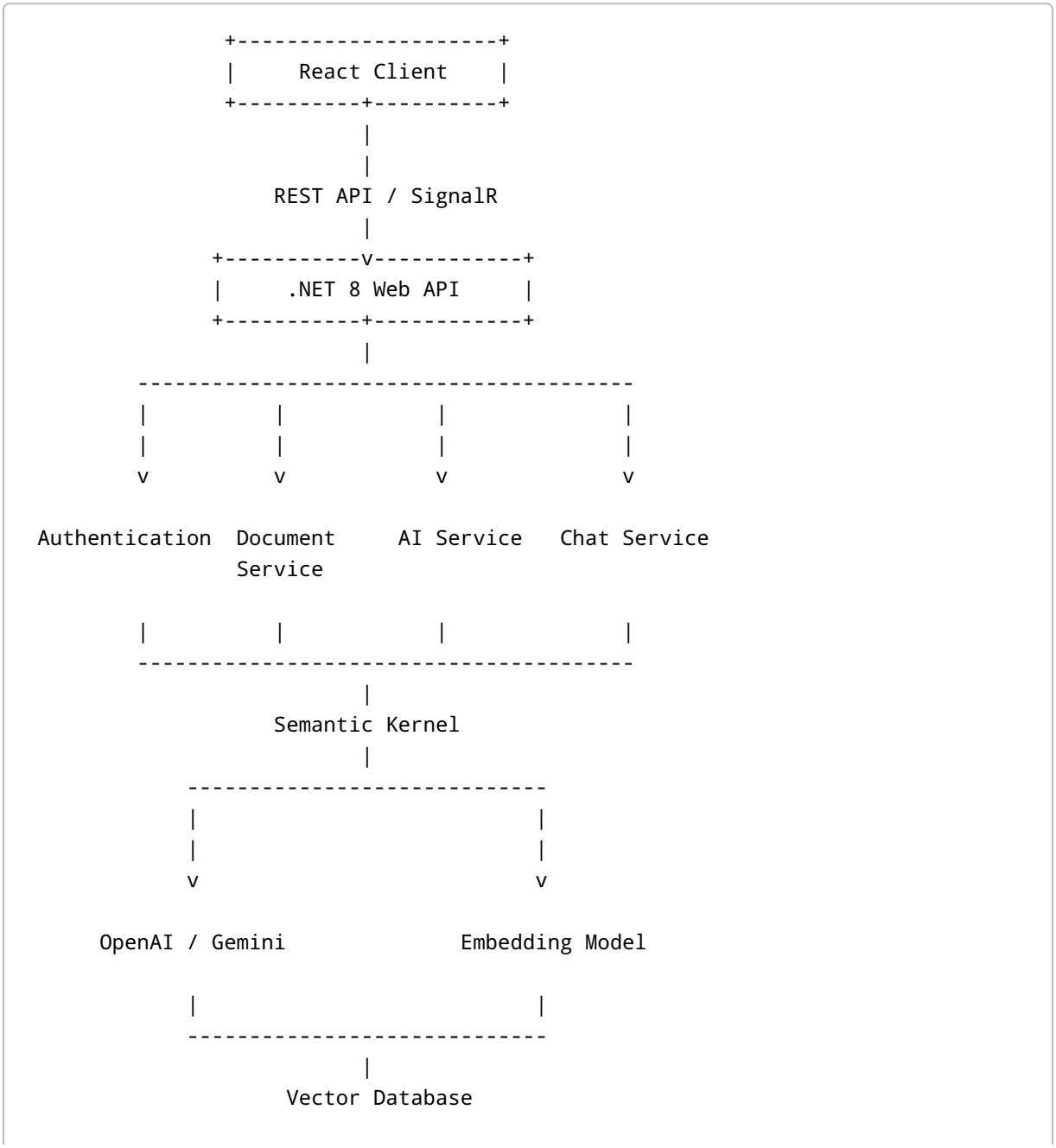
Storage

- Local Storage
- Azure Blob Storage
- AWS S3
- MinIO

Frontend

- React
- Next.js
- Tailwind CSS
- Shadcn UI

Overall Architecture



```
                (Pinecone / Milvus)
                          |
                          |
                PostgreSQL Database
```

High Level Flow

```
User

↓

Login

↓

Dashboard

↓

Upload PDF / DOCX

↓

Extract Text

↓

Split into Chunks

↓

Generate Embeddings

↓

Store Embeddings

↓

User asks Question

↓

Convert Question → Embedding
```

```
↓

Similarity Search

↓

Top Relevant Chunks

↓

Semantic Kernel Prompt

↓

OpenAI/Gemini

↓

Answer

↓

Display References
```

Folder Structure

```
src

├── DocumentAssistant.API
│
├── DocumentAssistant.Application
│
├── DocumentAssistant.Domain
│
├── DocumentAssistant.Infrastructure
│
├── DocumentAssistant.Persistence
│
├── DocumentAssistant.SemanticKernel
│
├── DocumentAssistant.VectorStore
│
```

```
├── DocumentAssistant.Shared
│
└── DocumentAssistant.Tests
```

Clean Architecture

```
Presentation

↓

Application

↓

Domain

↓

Infrastructure

↓

External Services
```

Modules

Authentication

- Register
- Login
- JWT
- Refresh Token
- Roles
- Permissions

User Module

```
Create User

Update User

Delete User

Profile

Settings
```

Document Module

```
Upload

Delete

Rename

Download

List Documents

View Metadata

View Pages
```

AI Module

```
Embedding Generation

Prompt Builder

Semantic Kernel Plugins

Context Window

Memory
```

```
Prompt Templates

Response Formatter
```

Vector Module

```
Store Embeddings

Delete Embeddings

Search Similar Chunks

Metadata Filter

Namespace Support
```

Document Processing Pipeline

```
PDF

↓

Read File

↓

Extract Text

↓

Clean Text

↓

Remove Empty Spaces

↓

Split Paragraphs

↓
```

```
Chunking

↓

Overlap

↓

Embedding

↓

Vector Database
```

Chunk Strategy

```
Chunk Size

500 Tokens

Overlap

100 Tokens
```

Example

```
Chunk 1

Page 1

Words 1-500

Chunk 2

Words 401-900

Chunk 3

Words 801-1300
```

Database Tables

Users

```
Id

Name

Email

PasswordHash

CreatedAt
```

Documents

```
Id

UserId

Name

FileType

StoragePath

Status

CreatedAt
```

Chunks

```
Id

DocumentId

ChunkIndex

Text
```

```
EmbeddingId

PageNumber
```

Conversations

```
Id

UserId

Title

CreatedAt
```

Messages

```
Id

ConversationId

Role

Content

CreatedAt
```

Pinecone Structure

```
Namespace

UserId

↓

DocumentId

↓
```

```
ChunkId

↓

Embedding
```

Metadata

```
UserId

DocumentId

Page

Chunk

Filename

CreatedDate
```

Semantic Kernel Flow

```
Question

↓

Generate Embedding

↓

Vector Search

↓

Top 5 Chunks

↓

Build Prompt

↓
```

```
OpenAI

↓

Answer

↓

Citation
```

Prompt Template

```
You are an AI assistant.

Use only the provided context.

If the answer is unavailable, say:

"I couldn't find that information in the uploaded documents."

Context

{{Context}}

Question

{{Question}}
```

Chat Flow

```
Ask Question

↓

Embedding

↓

Pinecone Search
```

```
↓

Top 5 Results

↓

Semantic Kernel

↓

LLM

↓

Streaming Answer

↓

Save Chat
```

API Endpoints

Authentication

```
POST /api/auth/register

POST /api/auth/login

POST /api/auth/refresh
```

Documents

```
POST /api/documents/upload

GET /api/documents

GET /api/documents/{id}

DELETE /api/documents/{id}
```

Embeddings

```
POST /api/embeddings/create

DELETE /api/embeddings/{documentId}
```

Chat

```
POST /api/chat/ask

GET /api/chat/history

GET /api/chat/{conversationId}
```

Admin

```
GET /api/users

GET /api/statistics
```

Security

- JWT Authentication
- Refresh Tokens
- HTTPS
- File Validation
- Virus Scan
- Rate Limiting
- API Keys in Secret Manager
- Prompt Injection Protection
- SQL Injection Protection
- XSS Protection

Logging

Serilog

```
Request Logs

Error Logs
```

```
AI Response Time

Embedding Time

Upload Time

Token Usage
```

Caching

Redis

```
Embedding Cache

Prompt Cache

User Cache

Document Cache
```

Background Jobs

```
Upload

↓

Queue

↓

Extract Text

↓

Generate Embeddings

↓

Store Vectors
```

```
↓

Notify User
```

Sequence Diagram

```
User

↓

Upload File

↓

API

↓

Storage

↓

Text Extraction

↓

Chunking

↓

Embedding

↓

Pinecone

↓

Completed
```

Chat Sequence

```
User

↓

Question

↓

Embedding

↓

Similarity Search

↓

Semantic Kernel

↓

GPT/Gemini

↓

Streaming

↓

Response
```

AI Features

- RAG Search
- Semantic Search
- Conversation Memory
- Multi-turn Chat
- Citation Support
- Source References
- Confidence Score
- Streaming Responses
- Multi-document Search

- Document Summarization
- Keyword Search
- Hybrid Search
- OCR Support
- Markdown Output
- PDF Export
- Prompt Templates
- AI Plugins
- Tool Calling

Future Features

- Image Understanding
- OCR for Scanned PDFs
- Voice Chat
- Speech-to-Text
- Multi-language Search
- Agentic AI Workflows
- Workflow Automation
- Email Document Analysis
- Knowledge Graph Integration
- Microsoft Teams Bot
- Slack Bot
- WhatsApp Bot
- Mobile Application

Deployment Architecture

```
React Frontend

↓

NGINX

↓

.NET 8 API

↓

Redis
```

```
↓

PostgreSQL

↓

Pinecone

↓

OpenAI / Gemini

↓

Blob Storage
```

Recommended Tech Stack

Layer	Technology
Frontend	React + Next.js + Tailwind CSS
Backend	.NET 8 Web API
AI Orchestration	Semantic Kernel
LLM	OpenAI GPT-4.1 / GPT-4o or Gemini 2.x
Embeddings	text-embedding-3-small / Gemini Embeddings
Vector Database	Pinecone (Cloud) or Milvus (Self-hosted)
Database	PostgreSQL
Cache	Redis
File Storage	Azure Blob Storage / AWS S3 / MinIO
Authentication	JWT + Refresh Tokens
Logging	Serilog
Background Jobs	Hangfire
Deployment	Docker + Kubernetes + NGINX

End-to-End Request Lifecycle

```
1. User signs in
2. Uploads one or more PDF/DOCX files
3. Files are stored in object storage
4. Background worker extracts text
5. Text is cleaned and chunked
6. Embeddings are generated using OpenAI/Gemini
7. Embeddings and metadata are stored in Pinecone/Milvus
8. User asks a question
9. Query embedding is generated
10. Vector similarity search retrieves the most relevant chunks
11. Semantic Kernel constructs the prompt with retrieved context
12. LLM generates an answer using only retrieved knowledge
13. Response streams to the UI with cited document/page references
14. Conversation history is persisted for future context
```

Recommended Development Phases

Phase 1

- Authentication
- User Management
- File Upload
- PostgreSQL Setup

Phase 2

- PDF/DOCX Parsing
- Chunking
- Embedding Generation
- Pinecone Integration

Phase 3

- Semantic Kernel Integration
- RAG Pipeline
- Chat Interface
- Streaming Responses

Phase 4

- Conversation History
- Citations

- Redis Cache
- Hangfire Background Jobs

Phase 5

- OCR
- Multi-document Search
- Admin Dashboard
- Docker & Kubernetes Deployment
- Monitoring and Performance Optimization

This architecture follows modern enterprise practices (Clean Architecture + CQRS + Semantic Kernel + Retrieval-Augmented Generation) and is designed to scale from a single-user proof of concept to a production-ready, cloud-native document assistant.